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TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers

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Abstract

The overcrowding of the emergency departments presents a major challenge, exacerbated by an aging population and increasing complex cases. Triage, which prioritizes patients according to severity, faces significant pressure due to limited resources and growing patient numbers. This study explores the integration of artificial intelligence (AI) to enhance the triage process. We developed and evaluated AI-based models, including Support Vector Machines (SVM), Random Forests (RF), Artificial Neural Networks (ANN), Gradient Boosting Machines (GBM), Linear Regression (LR), XGBoost and a stacking model, to predict patient triage levels using the Korean Triage and Acuity Scale (KTAS). Our findings demonstrate that AI models, particularly SVM and GBM, delivered the highest prediction accuracies of 79% and 78.7%, respectively. These models also performed well in terms of precision (80.04% and 75.36%), recall (71.94% and 73.36%), and F1-score (72.93% and 72.91%). The remaining algorithms still demonstrated strong predictive capabilities. The developed Stacking Model exhibited the highest prediction, achieving an accuracy of 80.05%, precision of 80.27%, recall of 73.26%, and an F1-score of 74.41%. This incremental gain in performance demonstrates the effectiveness of model stacking, as it capitalizes on the complementary strengths of different algorithms to enhance overall predictive accuracy.

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1. Introduction

The annual number of emergency room visits in Canada and the USA is 142 million [1]. Emergency departments triage patients according to severity to ensure that the most critical cases are dealt with quickly and safely. Emergency department overcrowding, a major challenge since the 1980s, is exacerbated by an aging population and an increase in

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complex cases. Many patients turn to emergency departments out of fear, anxiety, or difficulties accessing local care, often seeing them as the first option for medical attention or as a referral point to specialists. This trend highlights the urgent need to reorganize healthcare services to better regulate emergency admissions and promote ambulatory care [2].

Triage in emergency departments involves assessing the severity of patients' conditions and prioritizing them. It quickly identifies critical cases, ensuring immediate care and efficient allocation of medical resources, thereby reducing waiting times and optimizing care. Nurses responsible for triage face a heavy workload, worsened by the increasing number of patients. This can lead to delays in patient assessment and compromise the quality of care provided. A lack of resources, both personnel and equipment, further complicates the triage process [3].

Historically, triage in emergency departments evolved from informal systems based on healthcare professionals' clinical judgment to more structured and standardized methods. Systems like the Manchester Triage Scale (1990s) and the Canadian Triage and Acuity Scale (CTAS, 1999) are widely used today, improving consistency and reliability but still subject to human limitations such as judgment errors and inter-rater variation [6]. AI and machine learning advancements present new opportunities to automate and enhance these critical processes [7]. Emergency departments face challenges such as patient overload, limited resources, and time-critical requirements, exacerbated by seasonal variations, pandemics, and mass events leading to sudden patient influxes [8]. Improving triage efficiency and accuracy has become a priority for healthcare administrators and policymakers [9]. The demand for Machine Learning algorithms applied to emergency data has increased over the years [10]. Recently, the growing integration of artificial intelligence (AI) in the medical field has garnered increasing interest, particularly in emergency departments [4]. New AI approaches to patient triage have received particular attention due to urgent clinical needs.

Research shows AI can reduce waiting times and improve triage accuracy [4], while also standardizing assessments and reducing variations in clinical judgment [5]. This research project aims to develop practical AI-based solutions to facilitate the triage process for healthcare professionals. The results could benefit healthcare system managers, medical technology developers, and patients requiring emergency care. The study evaluates machine learning models to classify patients using the KTAS [11]. The solution processes medical data to classify patients into one of the 5 levels of KTAS using supervised learning algorithms, with the aim of increasing the trust of clinicians in AI-based models through precision and reliability [12].

The remainder of this paper is organized as follows. Section 2 reviews related studies. Section 3 details the dataset, data preparation, and models used. Section 4 presents the results and a summary of hyperparameters. Section 5 compares our findings with related work. Finally, Section 6 summarizes the key findings.

2. Literature review

Implementing effective triage systems in emergency departments is essential to ensure patients receive appropriate care promptly. Traditionally, these triage systems have been based on manual methods and standardized protocols, such as the Manchester Triage Scale and the Canadian Triage and Acuity Scale (CTAS) [6, 13]. However, although widely used, these methods are limited by inter-rater variability and the risk of human error. Integrating artificial intelligence (AI) and machine learning techniques into emergency triage promises to improve these processes by increasing accuracy and reducing response times. Recent advances in machine learning (ML) technologies have positioned them as a strong candidate to automate the triage decision-making process. This progress has led many researchers to use ML technologies to develop models that help predict patient hospitalization needs and prioritize them according to the intensity of care registered nurses provide during hospital emergency department (ED) triage. Many studies have demonstrated the superior performance of ML in predicting hospitalization and critical care outcomes compared to traditional triage models, particularly through nursing triage evaluation [14]. Kaldis et al. [15] explored the application of machine learning, particularly GBM, through AutoML, to predict hospital admissions in emergency departments. They used the MIMIC-IV-ED dataset and highlighted the role of key variables like acuity and waiting hours. The study demonstrated the potential of machine learning to improve hospital admission predictions. Raita et al. [16] investigated using machine learning models to predict patients' clinical outcomes during emergency department triage. They used clinical and administrative data, including vital signs and diagnoses, and demonstrated that machine learning models could improve prediction accuracy. Hong, Haimovich, and Taylor [17] developed machine learning algorithms to predict hospital admissions from emergency department triage data. Based on vital signs and diagnoses

Table 1. Summary of studies using various ML models and triage systems.

Author	ML Model Used	Triage System Used	Prediction Made	Calculated Metrics
V. Kaldis et al [15]	AutoML, Gradient Boosting Machines (GBM)	Not specified	Hospital admissions	AUC
Y. Raita et al [16].	Lasso regression, random forest, gradient boosted decision tree, deep neural network	Emergency Severity Index (ESI)	ICU admission or in-hospital death	AUC, net benefit
Woo Suk Hong et al [17].	Logistic regression, XGBoost, deep neural networks	Emergency Severity Index (ESI)	Hospital admission	AUC
G. Feretzakis et al [19].	Random Forest, Logistic Regression, Naive Bayes, Support Vector Machine (SVM), Decision Tree, Neural Network	Not specified (clinical features used)	Hospital Admission	F-measure, ROC Area
Yu et al [20].	Logistic Regression, Random Forest, Deep Neural Network	Initial Nursing Assessment, KTAS, SOFA	ICU admission, ER death	AUC
Yun et al [21].	XGBoost, Deep Neural Network	Korean Triage and Acuity Scale (KTAS)	Critical care outcome	AUC
Choi et al [22].	Logistic Regression, Random Forest, XGBoost	Korean Triage and Acuity Scale (KTAS)	KTAS level prediction	AUROC
Huilin Jiang et al [23].	Multinomial logistic regression, XGBoost, random forest, gradient-boosted decision tree	Emergency Severity Index (ESI)	Triage level	AUC, accuracy, macro-F1

from electronic medical records, their model showed superior performance in identifying patients requiring urgent hospitalization. However, they highlighted a lack of direct comparative studies between different machine learning algorithms for triage, making it difficult to determine the most effective approaches. Goto et al. [18] used machine learning to predict clinical outcomes for children in emergency departments, but their study was limited to a specific pediatric population, raising questions about the generalizability of the results. Feretzakis et al. [19] explored the application of machine learning models to predict hospital admissions from emergency department data, demonstrating overall strong results with several algorithms, particularly RF, which showed superior performance and potential for clinical decision-making support. Yu et al. [20] developed a machine learning and initial nursing assessment-based triage system for emergency departments to predict adverse clinical outcomes. The study evaluated various ML algorithms, including LR, RF, and Deep Neural Network (DNN), using both the full and low-dimensional (LD) datasets. The results demonstrated that the ML and initial nursing assessment-based triage system outperformed existing triage systems like KTAS and Sequential Organ Failure Assessment (SOFA). This study highlights the potential of integrating ML with nursing assessments to enhance the accuracy and efficiency of triage in emergency departments. Yun et al. [21] developed a machine-learning model to predict critical care outcomes for adult patients presenting to the emergency department using initial triage information. The study compared the performance of XGBoost and DNN models with the conventional KTAS model developed using LR. The study concluded that the XGBoost model outperformed the conventional triage model in predicting critical care outcomes, demonstrating the potential of machine learning to improve triage accuracy in emergency settings. Choi et al. [22] developed machine learning models to predict the KTAS levels for patients in the emergency department. The study employed LR, RF, and XGBoost models. The models demonstrated high predictive performance for both Random Forest and XGBoost, suggesting the potential of ML to enhance triage accuracy. Jiang et al. [23] developed machine learning models to support emergency department triage for patients with suspected cardiovascular disease. The study used data from 17,661 patients and evaluated various algorithms including multinomial logistic regression (MLR), XGBoost, RF, and gradient-boosted decision trees (GBDT). Key predictive variables included blood pressure, heart rate, oxygen saturation, and age. The study demonstrates the potential of machine learning to enhance the accuracy of triage in emergency settings for cardiovascular conditions.

Table 1 presents recent academic publications that have explored the use of ML techniques to predict triage outcomes.

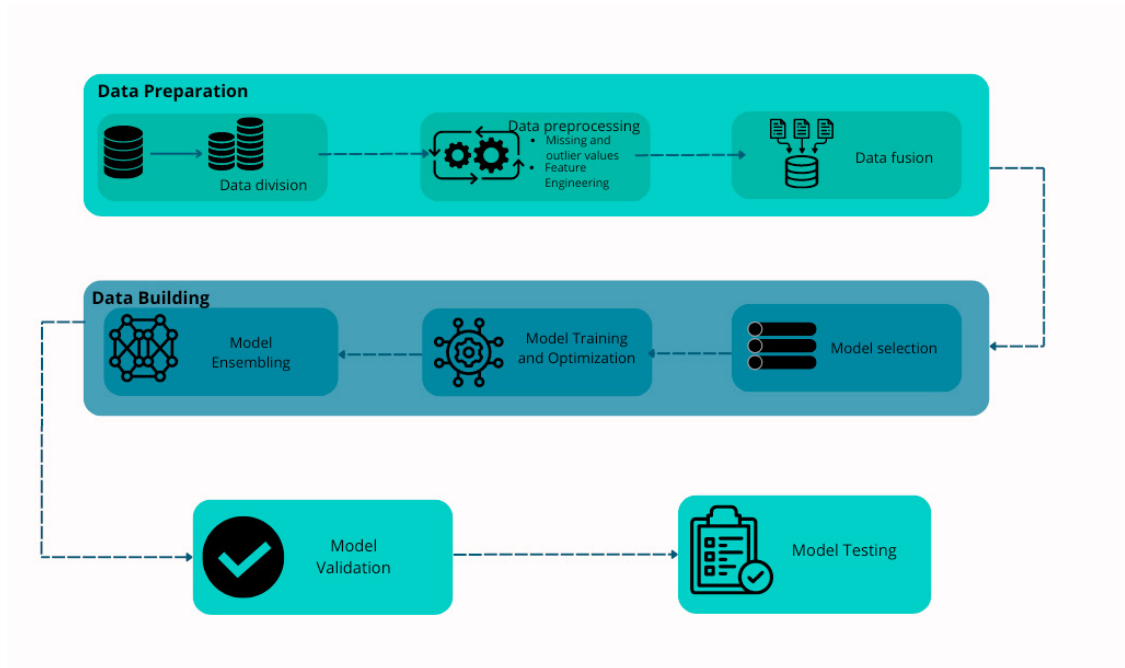


Fig. 1. Study design workflow.

3. Materials and Methods

3.1. Study design and setting

The data was obtained from Kaggle [24] which provided a well-structured dataset ideal for our analysis. The dataset ensured a diverse and representative sample that aligns with the study's objectives, with data points selected based on relevance, completeness, and diversity to provide a comprehensive view of the subject matter.

We conducted our analysis using Google Colab and Python version 3.10. Google Colab's cloud-based environment offered the flexibility and power needed for our computational tasks and the convenience of easy collaboration. The essential libraries for this project included NumPy for numerical operations, allowing us to handle large arrays and matrices of numerical data efficiently. We used pandas for data manipulation, which provided versatile data structures and functions designed to make data analysis fast and easy. For data visualization, we relied on Matplotlib and Seaborn, which enabled us to create informative and aesthetically pleasing graphs and plots, crucial for interpreting the results of our analysis. Lastly, we utilized scikit-learn for machine learning algorithms, offering a robust suite of tools for model training, evaluation, and validation.

These tools collectively facilitated a comprehensive and efficient workflow, enabling us to perform in-depth data analysis, generate meaningful visualizations, and develop robust predictive models. Integrating these libraries in Google Colab's environment allowed for seamless execution and sharing of our notebooks, enhancing our productivity and the quality of our research findings. The steps of our model are presented in Figure 1.

3.2. Data collection and processing

The data was collected from 1,267 systematically selected records of adult patients admitted to two emergency departments from October 2016 to September 2017. Twenty-four variables were evaluated, including chief complaints, initial vital signs recorded by nursing staff, and clinical outcomes. The true KTAS was determined by three triage experts: a certified emergency nurse, a KTAS provider and instructor, and a highly recommended nurse with outstanding emergency department experience and competence [11].

After collecting the data, we divided it into two datasets. Unnecessary attributes were carefully removed to streamline the analysis, while ensuring that the KTAS expert attribute was preserved as the target variable. Subsequently, we applied a series of preprocessing steps to each dataset, including handling missing values, addressing outliers, and performing feature engineering to enhance the quality and relevance of the data. This meticulous preprocessing ensured that both datasets were clean, consistent, and suitable for further analysis. For missing values, we identified three attributes 'Saturation', 'NRS_pain', and 'Diagnosis in ED' that were incomplete, and imputed them by filling in the most frequently occurring value for each case. Additionally, for outlier values, we replaced these with boundary values, moderating the effect of extreme data points without fully discarding them. In terms of feature engineering, categorical data initially represented as numerical values were transformed into meaningful categorical features, which enhanced the interpretability and utility of the data for modeling. Once preprocessing was complete, both datasets were merged into a comprehensive dataset. This combined dataset provided a holistic view of the patient's information, ready for further analysis and modeling.

3.3. Model Development

Since this research project focuses on a multiclass classification problem, specifically predicting the patient's emergency code. For our study, we evaluated six machine learning models: SVM, RF, ANN, GBM, LR, and XGBoost. Each of these models was selected for its distinct advantages and effectiveness in handling various types of data. SVMs are powerful for classification tasks, especially when data have clear separation margins, and they perform well in high-dimensional spaces [25]. LR, though a simpler model, remains highly interpretable and performs well in both binary and multiclass classification problems, serving as a robust baseline model. RF is an ensemble learning method that combines multiple decision trees to improve predictive performance and control overfitting, making it highly versatile and reliable [26]. ANNs, inspired by the human brain, are particularly useful for capturing complex patterns in data and are well-suited for tasks involving non-linear relationships due to their multi-layer structure. GBMs, known for their high predictive accuracy, build models in a stage-wise fashion and are particularly effective for both regression and classification tasks, especially when dealing with structured data [27]. XGBoost, an advanced implementation of gradient boosting, is recognized for its speed and superior performance in structured data scenarios, often outperforming other models due to its optimized boosting techniques. By comparing the performance of all six models, we aimed to determine which provided the best balance of accuracy, interpretability, and efficiency for predicting patient emergency codes. The inclusion of these diverse models ensured a comprehensive evaluation, allowing us to identify the most effective approach for this critical classification task. Training these models effectively requires careful tuning of hyperparameters, which is crucial for optimizing model performance. Hyperparameters control the learning process and can significantly affect models' accuracy, efficiency, and generalizability. In this study, we performed fine-tuning by testing various values for each hyperparameter across the six models. We employed techniques like grid search and random search to explore the hyperparameter space. Multiple combinations of hyperparameters were tested, and model performance was evaluated using cross-validation to avoid overfitting and ensure robust results. After comparing the outcomes across different hyperparameter configurations, we selected the values that yielded the best results in terms of accuracy, precision, recall, and F1-score. This detailed fine-tuning process was crucial in ensuring that each model was fully optimized for the multiclass classification task, allowing us to deliver high-performance models.

Our model pipeline was designed to ensure a consistent and robust data preparation and training process. The pipeline included several key steps: data preparation, data balancing, and model training. During the data preparation phase, we transformed categorical variables into numerical formats through encoding. To address class imbalance, we utilized SMOTE (Synthetic Minority Over-sampling Technique), which oversamples the minority class to prevent the models from being biased toward the majority class. While SMOTE is effective in balancing the dataset and preventing skewed model performance, it is important to consider its impact on the overall performance of the model. Synthetic data generated by techniques like SMOTE can sometimes lead to models that appear overly optimistic in their performance metrics, such as accuracy and F1-score. This is because the model learns from synthetic samples that may not fully represent the complexity of real-world, unseen data. As a result, while the model may perform well on the training and validation sets, it may struggle to generalize effectively when applied to new, unseen datasets. Despite this potential drawback, we chose to work with SMOTE because it provides a valuable approach to mitigating class imbalance, which is a common challenge in healthcare datasets where critical cases may be underrepresented. Without addressing this imbalance, the models could become biased toward the majority class, leading to suboptimal

predictions for minority class instances. By using SMOTE, we aimed to improve the model's ability to identify these rare cases, while remaining mindful of the need for further validation on real-world data to ensure the models' generalizability. The model training phase involved training various classification models (SVM, RF, ANN, GBM, LR, XGBoost) on the processed data. The pipeline ensured that the same transformations were consistently applied to both the training and test sets, ensuring that the models were trained and tested on identically processed data. Evaluating model performance is essential to understand how well the models generalize to new data. We used a comprehensive set of metrics to assess our models: accuracy, precision, recall, F1 score, AUC (area under the curve), ROC curve (receiver operating characteristic curve) and confusion matrix. Accuracy measures the overall correctness of the model, while precision evaluates the proportion of true positive predictions among all positive predictions. Recall assesses the proportion of true positive predictions among all actual positive cases, and the F1 score, as the harmonic mean of precision and recall, provides a balance between the two. The AUC represents the model's ability to distinguish between classes, and the ROC curve visualizes the trade-off between the true positive rate and the false positive rate. The confusion matrix summarizes the true positives, true negatives, false positives, and false negatives, providing a detailed breakdown of the model's performance [28, 29].

Finally, we combined our models using stacking, which integrates multiple models to improve overall performance. Stacking allows us to leverage the strengths of each individual model by combining their predictions through a meta-model, typically resulting in better generalization and performance than any single model on its own. By using stacking, we capitalize on the diversity of the underlying models, where each model may capture different patterns or nuances in the data. This synergy helps mitigate individual weaknesses, making the overall system more robust [30]. After combining the models, we calculated the metrics to evaluate the effectiveness of the stacked model and then tested it on new inputs to ensure its robustness and accuracy in practical applications. This approach allowed us to improve both prediction accuracy and model reliability. The decision to work with a stacking model is justified by its ability to reduce overfitting and enhance predictive performance by aggregating the insights from different algorithms. Stacking, by fusing models with varying strengths, helps produce a more balanced predictive system that is less prone to the biases or limitations of any single model, making it especially valuable for complex tasks like patient triage predictions.

4. Results

We evaluated various machine learning models, including SVM, RF, ANN, LR, XGBoost and GBM. A total of 6 models were trained and tested in the data set, yielding prediction precision as shown in Table 2. The results showed that GBM and SVM models delivered the highest prediction accuracies of 79% and 78.7%, respectively. These models performed well in precision (80.04% and 75.36%), recall (71.94% and 73.36%), and F1-score (72.93% and 72.91%).

While SVM and GBM achieved the highest accuracies, the remaining models still delivered commendable performance metrics, showcasing their effectiveness in predicting triage patient outcomes. The close accuracy scores of the other models highlight the overall efficacy of our model selection and parameter tuning process, underscoring the reliability and versatility of various machine learning approaches in handling complex classification tasks.

After testing multiple combinations, we developed a Stacking-Model that combining the best three-performed models. This ensemble model demonstrated a slight improvement in performance compared to the base models as shown in Table 3. The Stacking-Model's ability to leverage the strengths of each contributing model resulted in a more robust and accurate prediction, showcasing its potential advantage over standalone models. This minor performance boost highlights the efficacy of model stacking in enhancing predictive accuracy by combining the complementary strengths of different algorithms.

5. Discussion

Among the models tested in this study, the stacking model that combined SVM, GBM, and LR outperformed individual models. It achieved an accuracy of 80.05%, surpassing MLP, RF, XGBoost, and GBDT models from [23] by 5.75%, 5.55%, 1.55%, and 3.35%, respectively. Furthermore, the AUROC of our stacking model reached 93%, outperforming the LR, RF, and XGBoost models in [22] by 30%, 10%, and 10%, respectively, as detailed in our experiments notebook in [31].

Table 2. Performance metrics of various models.

Model	Hyperparameters	Accuracy	Precision	Recall	F1-score	AUC
SVM	kernel: 'rbf', C: 10, gamma: 0.01	78.74%	75.36%	73.36%	72.91%	0.9308
RF	n_estimators: 100, max_depth: 10, min_samples_split: 5, min_samples_leaf: 2, bootstrap: True, random_state: 42	76.90%	73.81%	73.83%	72.68%	0.9059
ANN	hidden_layer_sizes: (50, 25), activation: 'relu', solver: 'adam', alpha: 0.001, learning_rate: 'adaptive', max_iter: 500, random_state: 42, early_stopping: True, validation_fraction: 0.1	75.06%	71.51%	70.87%	69.67%	0.92
GBM	n_estimators: 100, learning_rate: 0.1, max_depth: 3, random_state: 42	79.00%	80.04%	71.94%	72.93%	0.9203
LR	penalty: 'elasticnet', l1_ratio: 0.5, C: 1, solver: 'saga', max_iter: 200	75.85%	77.90%	72.60%	74.14%	0.9068
XGBoost	n_estimators: 100, learning_rate: 0.1, max_depth: 3, random_state: 42	76.90%	76.14%	70.63%	72.05%	0.9199

Table 3. Performance metrics of the Stacking-Model.

Model	Base models	Accuracy	Precision	Recall	F1-score	AUC
Stacking-Model	SVM, GBM, LR	80.05%	80.27%	73.26%	74.41%	0.9298

Most existing research as noted in our literature review, focuses on predicting patient outcomes, but this study highlights the importance of accurate triage level predictions, which directly impact those outcomes. Precise triage helps identify high-risk patients early, enabling timely interventions, reducing complications, and improving survival rates. Errors in triage, such as over- or under-triage, can lead to delays in treatment and worsened patient conditions as noted by [11]. Therefore, improving triage accuracy through machine learning enhances emergency department operations and strengthens overall patient care quality by minimizing these misclassification errors.

6. Conclusion

This study highlights the potential of AI to enhance emergency department triage by improving the accuracy and efficiency of patient severity assessments through advanced machine learning models. Despite the promising results, a key limitation is the reliance on a single dataset, which may hinder the generalizability of the models to other clinical environments with varying demographics, seasonal factors, and hospital contexts. To address these limitations, future work will involve expanding the dataset to incorporate diverse clinical conditions, demographic variations, and seasonal patterns. This will improve the robustness and applicability of the models across different emergency department scenarios. Additionally, validating the models on new datasets will be crucial to ensuring their performance and adaptability in different healthcare settings. By increasing the diversity of data and testing the models in real-world environments, future research aims to strengthen the generalizability and reliability of AI-driven triage systems, ultimately improving patient outcomes and resource management in emergency care.

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